

Probability & Statistics (MT-2005)

Descriptive Statistics

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Numerical Descriptive Statistics

Numerical Descriptive Statistics: Key Points

- **Purpose of Numerical Measures:**

- Convey a mental image of data, phenomena, or objects.
- Provide a more concise description compared to graphical methods.

- **Limitations of Graphical Methods:**

- Inappropriate for statistical inference.
- Difficult to compare sample and population frequency histograms.

- **Data Reduction:**

- Reduces the number of entries describing a dataset.
- Comes with progressive loss of information.
- Requires identifying the most important characteristics of the distribution.

- **Focus of Numerical Descriptive Statistics:**

- Summarize main features of a dataset.
- Provide insights into:
 - Central tendency.
 - Variability.
 - Position.
 - Shape of data.

Measures of Central Tendency: Arithmetic Mean

Measures of Central Tendency: Arithmetic Mean

Measures of central tendency provide a single value that summarizes or represents the central point of a dataset. These measures help describe the overall characteristics of data and are widely used in statistics, data analysis, and research.

- Mean
- Median
- Mode

Ungrouped Data For a dataset $x_1, x_2, \dots, x_n$:

$$\text{Mean} = \bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

where n is the number of observations.

Grouped Data

$$\text{Mean} = \bar{x} = \frac{\sum_{i=1}^k f_i x_i}{\sum_{i=1}^k f_i}$$

where f_i is the frequency of the i -th class, x_i is the midpoint of the i -th class, and k is the number of classes.

Properties of Arithmetic Mean

- The sum of the deviations of the observations from their mean is equal to zero, i.e., $\sum_{i=1}^n (x_i - \bar{x}) = 0$

Statement: Given a dataset with n observations $x_1, x_2, \dots, x_n$, the mean $\bar{x}$ is defined as:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Proof: The Sum of Deviations from the Mean

Proof:

- Start with the sum of deviations:

$$\sum_{i=1}^n (x_i - \bar{x})$$

- Expand the expression inside the summation:

$$\sum_{i=1}^n (x_i - \bar{x}) = \sum_{i=1}^n x_i - \sum_{i=1}^n \bar{x}$$

- Since $\bar{x}$ is a constant:

$$\sum_{i=1}^n \bar{x} = n\bar{x}$$

- Substituting and simplifying:

$$\sum_{i=1}^n (x_i - \bar{x}) = \sum_{i=1}^n x_i - n\bar{x} = \sum_{i=1}^n x_i - \sum_{i=1}^n x_i = 0 \quad \text{proved}$$

The Sum of Squared Deviations from the Mean

- The Sum of Squared Deviations from the Mean is Minimum.

Statement: We aim to prove that the sum of squared deviations of x_i from their mean $\bar{x}$ is a minimum:

$$\sum_{i=1}^n (x_i - \bar{x})^2 \leq \sum_{i=1}^n (x_i - a)^2$$

where a is any arbitrary value other than $\bar{x}$.

Proof: The Sum of Squared Deviations from the Mean

- Start with the arbitrary value a :

$$\sum_{i=1}^n (x_i - a)^2$$

- Expand the squared terms:

$$\sum_{i=1}^n (x_i - a)^2 = \sum_{i=1}^n x_i^2 - 2a \sum_{i=1}^n x_i + na^2$$

Proof cont'd...

- The sum of squared deviations from the mean $\bar{x}$:

$$\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2$$

- Subtract the squared deviations from the mean from those of the arbitrary value a :

$$\sum_{i=1}^n (x_i - a)^2 - \sum_{i=1}^n (x_i - \bar{x})^2 = n(a - \bar{x})^2$$

- Since $n(a - \bar{x})^2 \geq 0$ for all a :

$$\sum_{i=1}^n (x_i - a)^2 \geq \sum_{i=1}^n (x_i - \bar{x})^2 \quad \text{Proved}$$

Transformation of Mean under Scaling and Shifting

For a dataset $\{x_1, x_2, \dots, x_n\}$ with mean $\bar{x}$, if each observation is transformed as:

$$x'_i = k \cdot x_i + c,$$

the new mean becomes:

$$\bar{x}' = k \cdot \bar{x} + c.$$

Proof:

- Start with the original dataset and the new dataset transformed by scaling and shifting.
- Substitute the transformed observations into the mean formula:

$$\bar{x}' = \frac{1}{n} \sum_{i=1}^n (k \cdot x_i + c)$$

- Simplify the summation and use the properties of the sum:

$$\bar{x}' = k \cdot \bar{x} + c$$

Characteristics of Arithmetic Mean

- **Mean as Projection onto the Span of Ones.** The arithmetic mean can be interpreted as the projection of a data vector onto the span of the vector of ones. (**Think!**)
- It is sensitive to extreme values (outliers). If there is a very large or very small number in the data set, it can significantly change the mean.
- The mean is best used when the data is symmetrically distributed or when all data points have equal importance and outliers are minimal.

Measures of Central Tendency: Median

Median: Ungrouped Data

Ungrouped Data: Arrange the data in ascending order:

- If n is odd, the median is the middle value.
- If n is even, the median is the average of the two middle values.

Median: Grouped Data

Grouped Data: The median is calculated using the formula:

$$\text{Median} = L + \left(\frac{\frac{N}{2} - CF}{f_m} \right) \cdot w$$

where:

- L : Lower boundary of the median class
- N : Total frequency
- CF : Cumulative frequency before the median class
- f_m : Frequency of the median class
- w : Width of the median class

Characteristics of Median:

- It is not affected by outliers or extreme values in the data set.
- The median is especially useful when the data is skewed or contains outliers because it better represents the "typical" value.

Measures of Central Tendency: Mode

Mode: Ungrouped Data

Ungrouped Data: The mode is the most frequently occurring value in the dataset.

Mode: Grouped Data

$$\text{Mode} = L + \left(\frac{f_m - f_{m-1}}{(f_m - f_{m-1}) + (f_m - f_{m+1})} \right) \cdot w$$

where:

- L : Lower boundary of the modal class
- f_m : Frequency of the modal class
- f_{m-1} : Frequency of the class before the modal class
- f_{m+1} : Frequency of the class after the modal class
- w : Width of the modal class

Characteristics of Mode:

- The mode is not affected by outliers since it is based on frequency.
- It is especially useful for categorical data (e.g., color, gender) where you want to know which category is most common.

Finding Mean, Median, and Mode for Grouped Data

Example: Finding Mean, Median, and Mode for Grouped Data

We have the following data on the scores of 40 students in a class:

{50, 55, 58, 60, 63, 65, 68, 70, 72, 73, 75, 78, 80, 82, 85, 87, 88, 90,
92, 95, 96, 97, 100, 102, 105, 108, 110, 113, 115,
118, 120, 123, 125, 128, 130, 135, 138, 140, 142, 145}

Step 1: Number of Classes Using Sturges' Rule

Sturges' rule gives the number of classes (k) for grouped data:

$$k = 1 + 3.322 \log_{10} n$$

where n is the number of data points.

For our dataset:

$$n = 40 \Rightarrow k = 1 + 3.322 \log_{10} 40 = 1 + 3.322 \times 1.602 = 1 + 5.318 = 6.318$$

So, we round it up to $k = 7$ classes.

Step 2: Calculate Class Width

Class width (w) is calculated as:

$$w = \frac{\text{Range}}{k}$$

where the range is the difference between the maximum and minimum values in the dataset.

$$\text{Range} = 145 - 50 = 95$$

$$w = \frac{95}{7} \approx 13.57$$

Example cont'd...

We round the class width to the nearest whole number, so $w = 14$.

Step 3: Define Class Limits and Boundaries

The first class starts at 50, and the class width is 14. So the class limits and boundaries are:

Class Limits	Class Boundaries
50 – 63	49.5 – 63.5
64 – 77	63.5 – 77.5
78 – 91	77.5 – 91.5
92 – 105	91.5 – 105.5
106 – 119	105.5 – 119.5
120 – 133	119.5 – 133.5
134 – 147	133.5 – 147.5

Step 4: Frequency Distribution Table

Now, let's group the data and create the frequency distribution table. For simplicity, we provide the frequencies for each class:

Class	Frequency(f)	Midpoint(x)	Cumulative Frequency(CF)
50 – 63	5	56.5	5
64 – 77	7	70.5	12
78 – 91	9	84.5	21
92 – 105	6	98.5	27
106 – 119	5	112.5	32
120 – 133	5	126.5	37
134 – 147	3	140.5	40

Example cont'd...

Mean: The mean is calculated as:

$$\bar{x} = \frac{\sum(f \cdot x)}{\sum f}$$

Where f is the frequency and x is the midpoint of the class.

$$\bar{x} = \frac{(5 \cdot 56.5) + (7 \cdot 70.5) + (9 \cdot 84.5) + (6 \cdot 98.5) + (5 \cdot 112.5) + (5 \cdot 126.5)}{5 + 7 + 9 + 6 + 5 + 5 + 3}$$

$$\bar{x} = \frac{282.5 + 493.5 + 760.5 + 591.0 + 562.5 + 632.5 + 421.5}{40} = \frac{4744}{40} = 118.6$$

$$\text{Mean} = 118.6$$

Example cont'd...

Median: The median class is the class where the cumulative frequency reaches or exceeds half the total frequency.

The median class is Class 3 (78 - 91), (the first class after CF become greater than $N/2 = 20$). The median is calculated using the formula:

$$\text{Median} = L + \left(\frac{\frac{N}{2} - CF}{f_m} \right) \cdot w$$

where:

- $L = 77.5$ (lower boundary of the median class),
- $N = 40$ (total frequency),
- $CF = 12$ (cumulative frequency before the median class),
- $f_m = 9$ (frequency of the median class),
- $w = 14$ (class width).

Example cont'd...

$$\text{Median} = 77.5 + \left(\frac{\frac{40}{2} - 12}{9} \right) \cdot 14 = 77.5 + \left(\frac{20 - 12}{9} \right) \cdot 14$$

$$\text{Median} = 77.5 + \left(\frac{8}{9} \right) \cdot 14 = 77.5 + 12.44 = 89.94$$

$$\text{Median} \approx 90$$

Example cont'd...

Modal Class: The modal class is the class with the highest frequency, which is the class 78 - 91 with frequency $f_1 = 9$.

Formula for Mode:

$$\text{Mode} = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \cdot h$$

where:

- L : Lower class boundary of the modal class,
- f_1 : Frequency of the modal class,
- f_0 : Frequency of the class before the modal class,
- f_2 : Frequency of the class after the modal class,
- h : Width of the class.

Example cont'd...

Given Values:

- $L = 77.5$ (lower class boundary of the modal class),
- $f_1 = 9$ (frequency of the modal class),
- $f_0 = 7$ (frequency of the class before the modal class: 64 - 77),
- $f_2 = 6$ (frequency of the class after the modal class: 92 - 105),
- $h = 14$ (class width).

Substitute the values into the formula:

$$\text{Mode} = 77.5 + \left(\frac{9 - 7}{2 \times 9 - 7 - 6} \right) \cdot 14$$

Simplifying:

$$\text{Mode} = 77.5 + \left(\frac{2}{18 - 7} \right) \cdot 14 = 77.5 + \left(\frac{2}{11} \right) \cdot 14$$

$$\text{Mode} = 77.5 + 0.1818 \cdot 14 = 77.5 + 2.545 = 80.045$$

Thus, the **Mode** is approximately:

Example: Dataset Without Outliers

A server processes requests, and the response times (in milliseconds) for a given period are recorded as follows:

Response Times: 120 ms, 130 ms, 125 ms, 140 ms, 135 ms.

Mean:

$$\text{Mean} = \frac{120 + 130 + 125 + 140 + 135}{5} = 130 \text{ ms.}$$

Median:

Sorted Data: 120, 125, 130, 135, 140

$\implies$ Median = 130 ms.

Mode: No repeated value, so no mode.

Example: Dataset With Outliers

Consider a situation where one response time is exceptionally high:

Response Times: 120, 130, 125, 140, 135, 1000.

Mean:

$$\text{Mean} = \frac{120 + 130 + 125 + 140 + 135 + 1000}{6} \approx 275 \text{ ms.}$$

Median:

Sorted Data: 120, 125, 130, 135, 140, 1000

$$\Rightarrow \text{Median} = \frac{130 + 135}{2} = 132.5 \text{ ms.}$$

Mode: No repeated value, so no mode. **Interpretation:** The median remains relatively stable, as it is less sensitive to extreme values.

Other Types of Measures of Central Tendency

1. Geometric Mean

Definition: The n th root of the product of n values.

Formula:

$$\text{Geometric Mean} = \left(\prod_{i=1}^n x_i \right)^{\frac{1}{n}}$$

Example: For the dataset [1, 3, 9, 27]:

$$\text{Geometric Mean} = (1 \cdot 3 \cdot 9 \cdot 27)^{\frac{1}{4}} = \sqrt[4]{729} = 9$$

Characteristics:

- Useful for data with exponential growth (e.g., populations, compound interest).
- Less sensitive to extreme values than the arithmetic mean.
- Suitable for multiplicative relationships (e.g., rates of change).

2. Harmonic Mean

Definition: The reciprocal of the arithmetic mean of reciprocals.

Formula:

$$\text{Harmonic Mean} = \frac{n}{\sum_{i=1}^n \frac{1}{x_i}}$$

Example: For the dataset [2, 3, 4]:

$$\text{Harmonic Mean} = \frac{3}{\frac{1}{2} + \frac{1}{3} + \frac{1}{4}} = \frac{3}{\frac{13}{12}} = \frac{36}{13} \approx 2.77$$

Characteristics:

- Best for data representing rates, such as speed or efficiency.
- Highly influenced by small values.

3. Weighted Mean

Definition: Averages data points with different weights.

Formula:

$$\text{Weighted Mean} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}$$

Example: For the dataset [10, 20, 30] with weights [1, 2, 3]:

$$\text{Weighted Mean} = \frac{(10 \cdot 1) + (20 \cdot 2) + (30 \cdot 3)}{1 + 2 + 3} = \frac{140}{6} \approx 23.33$$

Characteristics:

- Useful when data points have different levels of importance or represent different sample sizes.
- Common in grading systems and survey data.

4. Trimmed Mean

Definition: Removes extreme values before calculating the mean.

Example: For the dataset $[1, 2, 3, 100]$, trim 25%:

$$\text{Trimmed Mean} = \frac{2 + 3}{2} = 2.5$$

Characteristics:

- Reduces the influence of outliers and extreme values.
- Useful in situations where data may contain errors or extreme values.

5. Quadratic Mean (Root Mean Square)

Definition: Square root of the mean of squared values.

Formula:

$$\text{Quadratic Mean} = \sqrt{\frac{\sum_{i=1}^n x_i^2}{n}}$$

Example: For the dataset [3, 4, 5]:

$$\text{Quadratic Mean} = \sqrt{\frac{3^2 + 4^2 + 5^2}{3}} = \sqrt{\frac{50}{3}} \approx 4.08$$

Characteristics:

- Useful when data represents squared values, such as deviations or in signal processing.
- Larger values can have more influence than smaller values.

6. Power Mean

Definition: A generalization of the mean.

Formula:

$$\text{Power Mean} = \left(\frac{\sum_{i=1}^n x_i^p}{n} \right)^{\frac{1}{p}}$$

Properties:

- For different values of p , the power mean interpolates between various means:
 - $p = 1$: The power mean becomes the **arithmetic mean**.
 - $p = 0$: The power mean approaches the **geometric mean**.
 - $p = -1$: The power mean becomes the **harmonic mean**.
 - $p = 2$: The power mean becomes the **quadratic mean (root mean square)**.
- When $p \rightarrow \infty$: The power mean approaches the **maximum value** of the dataset.
- When $p \rightarrow -\infty$: The power mean approaches the **minimum value** of the dataset.

Characteristics:

- Flexible mean that generalizes other means depending on the value of p .
- Useful for situations where different types of averaging are needed.

7. Midrange

Definition: The average of the minimum and maximum values in a dataset.

Formula:

$$\text{Midrange} = \frac{\text{Maximum Value} + \text{Minimum Value}}{2}$$

Characteristics:

- Not robust to outliers, as it only considers the minimum and maximum values.
- May not represent the data well if there are outliers.
- Useful for a quick estimate of central tendency.